

Rahul Manoj, PhD

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Curious and growth-oriented electrical engineer and research scientist with 7+ years of experience delivering technology solutions at the intersection of hardware, software, and modelling, with a strong background in applied R&D and cross-functional stakeholder collaboration.

KEY QUALIFICATIONS

- **Ultrasound Technologies:** System design, integration and algorithm development for A-Mode, B-Mode imaging, high frame rate plane wave imaging and Doppler flow imaging technologies.
- **Signal Processing & Instrumentation:** Hands-on experience in electronic system design and integration of sensors & transducers for biomedical & industrial usecases. Skilled in characterization, data acquisition and signal processing.
- **Prototyping & Laboratory Experiments:** Developed functional prototypes, experiment setups, and test rigs. Skilled in troubleshooting, measurement techniques, hardware-software integration of electro-mechanical systems.
- **Leadership, Collaboration & Technical Sales:** Initiated and led multi-institution technology collaborations by identifying opportunities, aligning stakeholder interests, and translating them into funded, deliverable projects.
- **Scientific Communication & IP:** Published high-impact journal articles, presented at international conferences, joint patent applications, joint grant applications, proposals for ethical approval of clinical studies.

PROFESSIONAL EXPERIENCE

Research Scientist | Division of Biomechanics

2024 to Present

Dept. of Structural Engineering, Norwegian University of Science & Technology, Trondheim, Norway

- Led the development of an end-to-end modelling pipeline integrating ultrasound data and physics-based models, delivering a validated decision-support solution in collaboration with clinicians for early-stage Heart Failure Diagnosis.
- Establish regulatory-aligned verification and validation frameworks (as per ASME V&V 40) to support translation of computational models into clinical practice.
- Contributing to strategic planning, new project ideation, proposal writing, and competitive grant applications.
- Technical sales & collaboration: Initiated and led multi-disciplinary collaborations by identifying opportunities, aligning interests, designing work packages, budgetary planning and translating them into potential grant applications, for deliverable technology development projects.

Researcher | Advanced Cardiovascular Technologies Laboratory

2017 to 2024

Dept. of Electrical Engineering, Indian Institute of Technology (IIT) Madras, Chennai, India

- Designed, built, and validated multi-modal instrumentation systems combining ultrasound, strain-gauge sensors, ECG, PPG and embedded electronics for various vascular health monitoring applications.
- Algorithm development for novel physiological biomarkers and improved estimation of clinically relevant biomarkers.
- Took ownership of system integration, testing, troubleshooting, verification, validation and characterization across multiple sensing prototypes measuring physiological signals from complex sensing environments.
- Hands-on wet lab experience in designing and building custom ex-vivo/in-vitro test rigs (semi-automated electromechanical systems) for verification and validation studies on excised animal blood vessels & silicon tubes.

Healthcare Technology Innovation Centre, IIT Madras Research Park, Chennai, India

- Co-designed and built a functional prototype of a modular and periodic weight measurement system for bedridden patients on hospital beds (Industry Collaborator: Stryker Corporation, India).
- Developed software and integrated electronics, sensors and actuators for an in-house designed automated cover slipper for large-format (6" X 8") histopathology human brain section slides.
- Co-drafted the medical device software lifecycle document for an in-house developed cardiovascular health assessment device ARTSENS® Plus as per IEC 62304 (Industry Collaborator: VASC RISK LLC, USA).

SKILLS & EXPERTISE

- Technology Development & Validation
- Programming: LabVIEW | MATLAB | Python
- Hardware Prototyping | Test & Measurement
- Physics & Data – Driven Modelling | AI Adoption
- Uncertainty Quantification | Sensitivity Analysis
- Embedded Programming | SPICE Simulations
- Electronics | PCB Design | Signal Processing
- Git | Linux | Windows | MS Office
- Critical Thinking | Problem Solving
- Collaboration | Project Management
- Scientific Communication | Patents

SELECTED ACHIEVEMENTS

- Authored six scientific articles in peer reviewed, high impact biomedical engineering journals and over 40 international conference proceedings on novel physiological algorithms and instrumentation for medical applications.
- Filed 5 patents in total as a co-inventor in India, EU and US. Granted: 1, Published: 4
- Won the popular prize for pitch deck Presentation for the *International Modelling Challenge: Digital Twin Builder for Health Incubator* by iSi Health KU Leuven, Belgium (June, 2025).

EDUCATION

(Certificates & education recognised by direktoratet for høyere utdanning og kompetanse, Norway)

- **M.S. & Ph.D.** in Electrical Engineering
(2017-2024) (Biomedical Instrumentation) **Indian Institute of Technology Madras (IIT M)**
Chennai, Tamil Nadu, India
- **B. Tech** in Electrical and Electronics Engineering
(2012-2016) **National Institute of Technology Calicut (NIT C)**
Kozhikode, Kerala, India

PERSONAL PROFILE

- Community Outreach: Member in technical bodies, organising workshops, reviewer for scientific journals.
- Hobbies & Interests: Swimming, outdoor activities & travelling, science blogging, electronics hobbyist.
- Languages: English (C2), Norwegian (A2/B1), Indian Languages: Malayalam (native), Hindi (B2), Tamil (A2)

REFERENCES

Prof. Leif Rune Hellevik
Professor, Department of Structural Engineering,
Norwegian University of Science & Technology (NTNU),
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Prof. Victorien Prot
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